Shot Estimation

Manav Seksaria

IIT Madras

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Introduction

- For current NISQ-era computer shot estimation remains a difficult problem.
- There has been work in the past analysing the theoretical convergence of specific circuits.[1], and statistical properties of qubits[2] to get better estimates. The general approach however has been trial-and-error[3]
- Here we will run a statistical analysis on various forms of noise, to then be able to estimate how many shots are required to obtain a result within known error bounds.

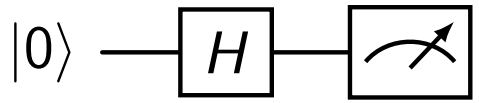
[1] Peruzzo, McClean, et al. 2014; Sweke, Wilde, et al. 2020.
[2] Burnett et al. 2019.
[3] Sung, Yao, et al. 2020; Brown, Goktas, et al. 2020.

Plan

- Establish a baseline for a single-qubit circuit for with one H gate and measurement.
- Add each source of noise individually to the baseline and plot the results
- Find patterns in variations from basline for each source of noise and make a prediction mechanism
- We expect to end with a formula for accurate shot estimation within known error bounds

Introduction

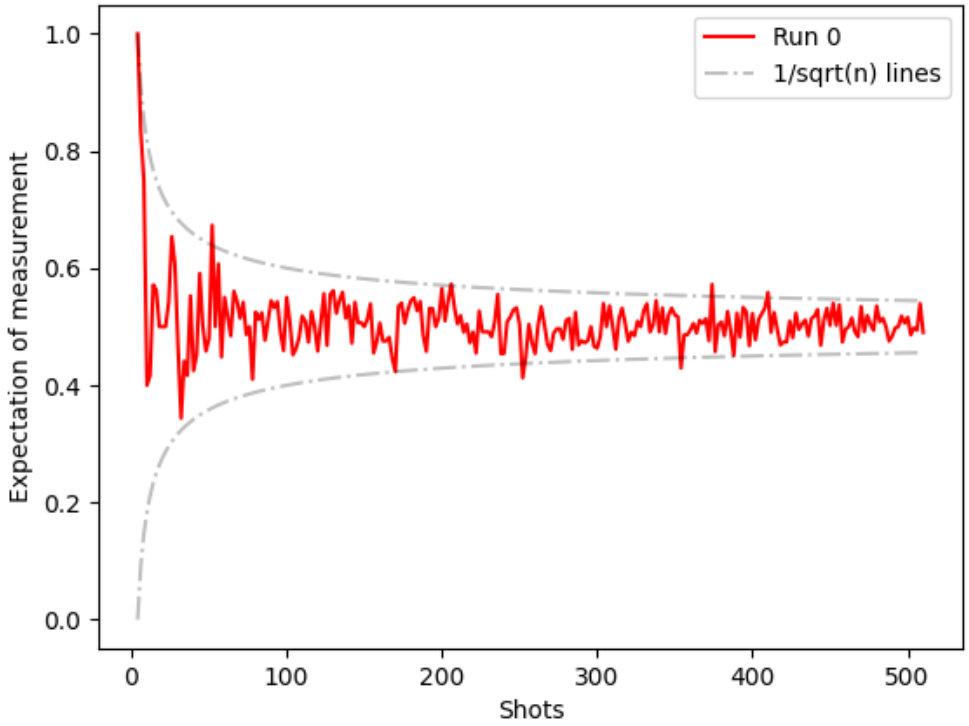
- Consider a single-qubit circuit with one H gate and measurement as our baseline.



- Over some n measurements we will obtain a distribution with probabilities of $0, 1$.
- We find the expectation of the measurement X as $\mu = Operatorname{E}[X] = \sum_{i=1}^{\infty} x_i p_i$ and plot μ vs n

Baseline

- Number of shots (2 to 512 shots)



- Width at any n decreases as the number of shots (n) increases, as $1/\sqrt{n}$.
- Compare this plot to noise models of isolated sources of noise.

Common sources of Noise

- **SPAM Noise** - State Preparation And Measurement errors arise either during the preparation of the qubit, or during measurement.
- **Relaxation Noise** - Qubits will decay to the ground state over time i.e $|1\rangle \rightarrow |0\rangle$.
- **Dephasing Noise** - Qubits will lose phase information overtime and go to an equal superposition of $|+\rangle$ and $|-\rangle$.
- **Gate Noise** - Errors in the application of gates single and multi-qubit gates
- **Crosstalk Noise** - Errors in the application of gates due to qubits being too close to each other.
- **Correlated Errors** - Other sources of noise that are not independent of each other.

Crosstalk and Correlated errors have no widely accepted models[4]

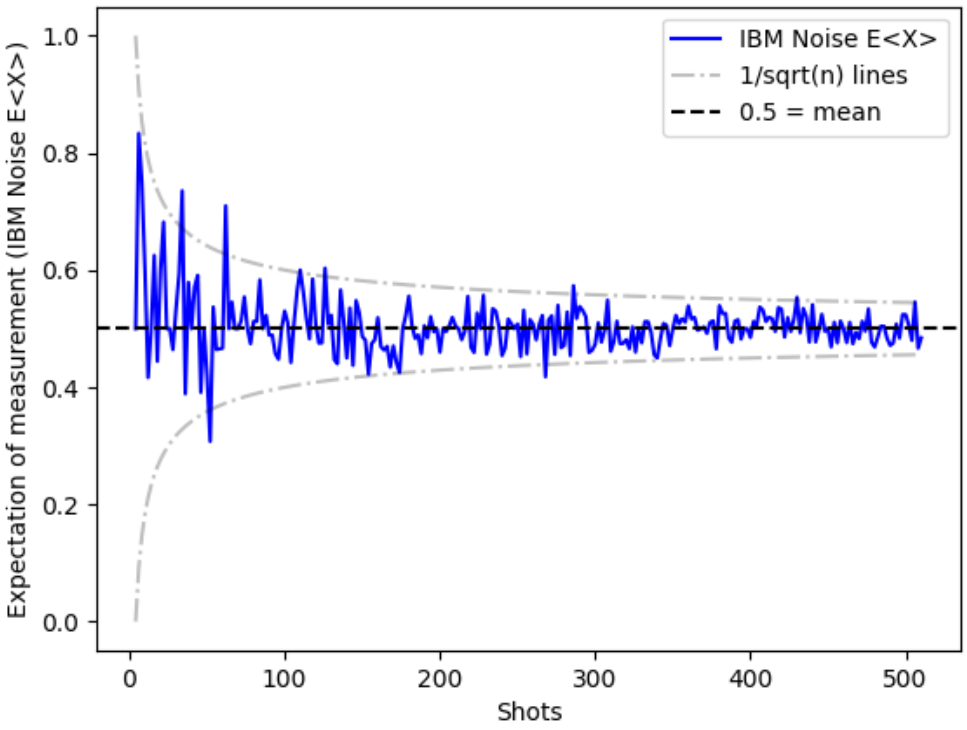
[4] Bravyi et al. 2018; Greenbaum and Dutton 2017.

SPAM Noise

- State Preparation And Measurement (SPAM) errors arise either during the preparation of the qubit, or during measurement.
- Noise simulators only allow us to simulate readout errors, not preparation errors.
  - We will only consider the measurement error in this case.
- Readout errors are made up of two components,
  - probability of a 0 read as a 1 i.e $p_{0\to 1}$,
  - probability of a 1 read as a 0 i.e $p_{1\to 0}$.

SPAM Noise

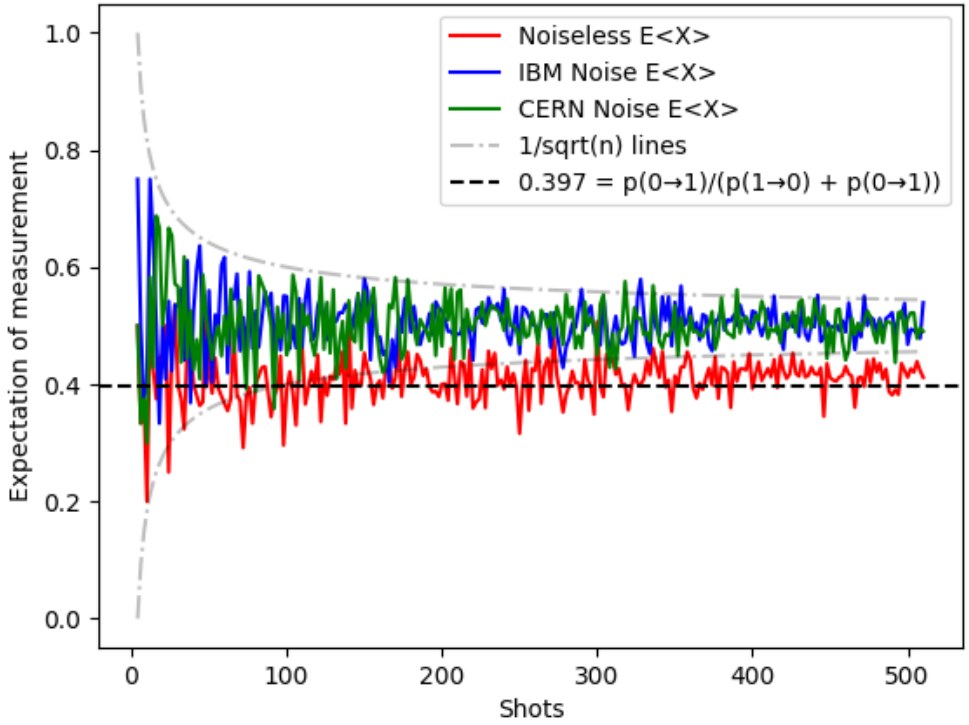
Plot for $p_{0\to1} = p_{1\to0} = 0.33$ on IBM's noise simulator with same baseline circuit.



The following slide has all plots on noise-free, IBM's noise, and CERN's noise simulator.

SPAM Noise

- Increase in width is not visually obvious, so we add asymmetric noise to IBM's simulator to visualise it's effects. for $p_{1\to 0} = 0.5$ and $p_{0\to 1} = 0.33$.



- New mean $\mu' = p_{0\to 1}/(p_{1\to 0} + p_{0\to 1})$ is shifted down when $p_{1\to 0} \neq p_{0\to 1}$.

Difference from mechanism so far

- Instead of taking mean/stdv of n shots, we take mean/stdv of windows of w shots
- New means for N windows of size w form new distribution $\bar{X}$.
- Ex, if we have results of 1024 shots, we would take say $N = 4$ windows of say width $w = 16$ shots each.

$$\text{Windows} = \{[0, 16], [16, 32], [32, 48], [48, 64]\}$$

- To get means of windows, $\mu_{w,1}, \mu_{w,2}, \mu_{w,3}, \mu_{w,4}$.
- New random variable $\bar{X} = \mu_{w,i}$ is now a normal distribution via central limit theorem as $w \to \infty$.

Difference from mechanism so far

- CLT Example: if $X \sim \text{Poisson}(\lambda)$, then $\frac{\bar{X}-\lambda}{\sqrt{\lambda}} \to N(0,1)$ as $\lambda \to \infty$
- So in our case $X \sim \text{Binomial}(n,p)$, then $\frac{\bar{X}-np}{\sqrt{np(1-p)}} \to N(0,1)$ as $n \to \infty$
- Our distribution was not originally normal, we created a new distribution $X \to \bar{X}$ to make it normal and therefore got fixed $\frac{1}{2}$ slope on a log(RSD) vs log(w) plot as we will see.

Applying CLT Mechanism

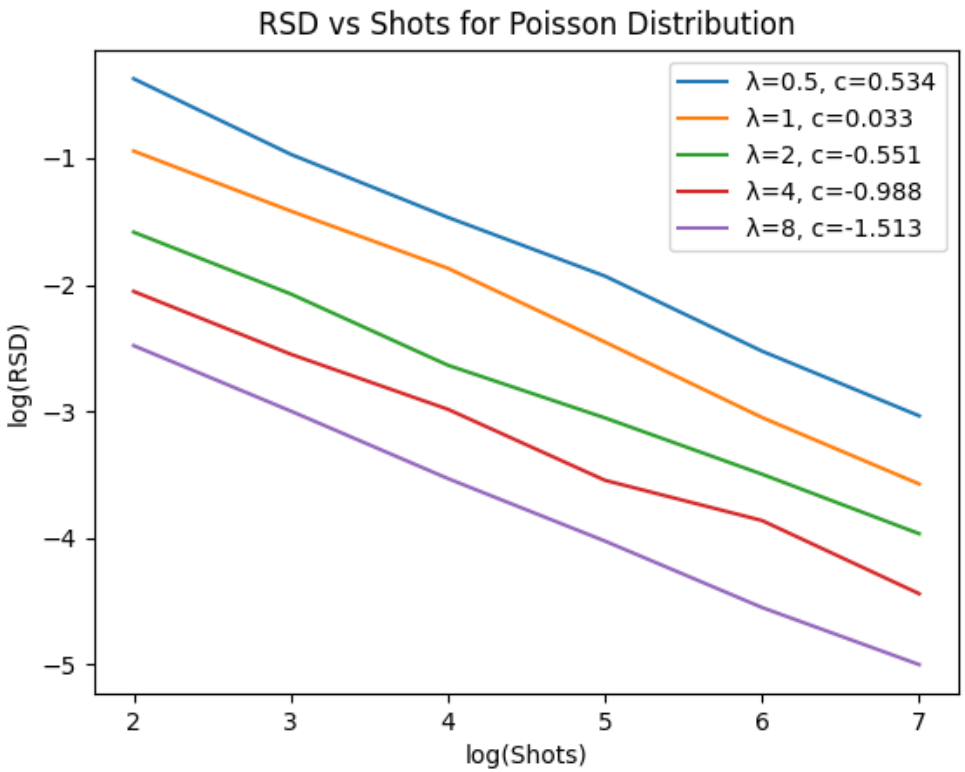
If we were to apply the CLT mechanism to the SPAM noise, and run the following experiment, we do the following procedure.

1. Run the circuit for 2^{15} shots (with $p_{0\to1} = p_{1\to0} = 0.33$)
2. Divide the results into first $N = 256$, windows of $w = 2^2$ shots
3. Calculate the mean of each window, to get a new distribution of 256 values which is normal
4. Calculate the mean, stdv of this new distribution
5. Repeat process for windows of size $w = \{2^3, 2^4, 2^5 \ldots 2^{11}\}$
6. Plot the log(RSD) vs log(w) for each window size [5]
7. We should see a fixed $\frac{1}{2}$ slope for each window size with varying intercepts

[5] Note here we're plotting against w and not n. Number of windows taken is fixed at $N = 256$

Applying CLT Mechanism (Poisson)

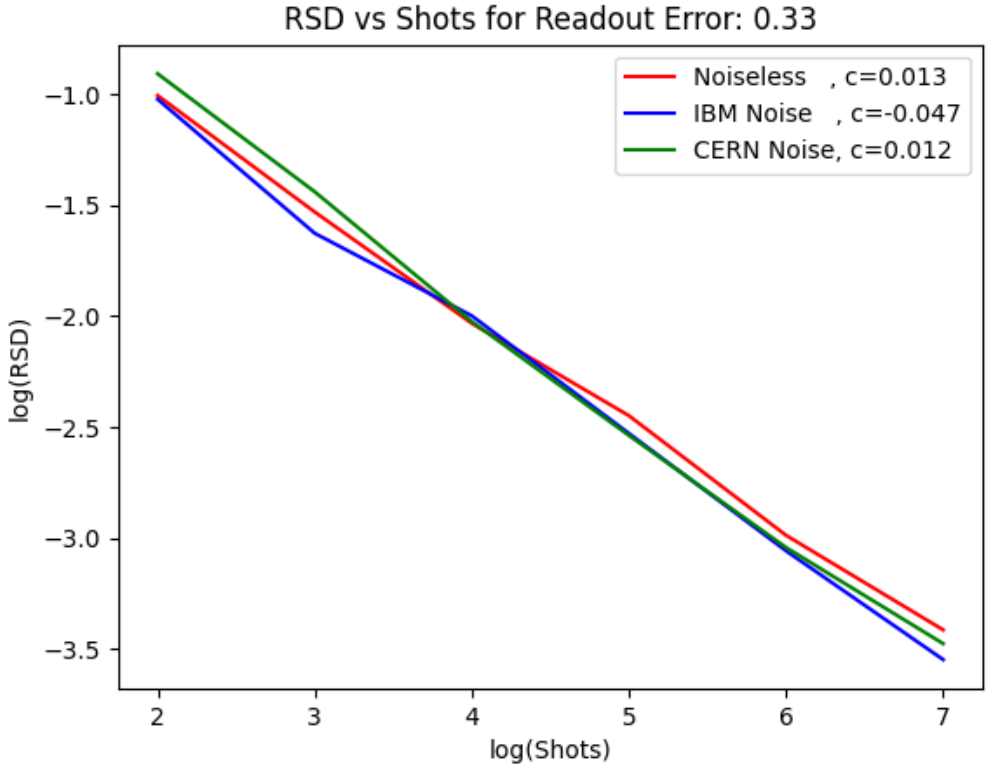
- CLT mechanism on Poisson distribution as an example
- $\lambda \in \{0.5, 1, 2, 4\}$, $n = 2^{15}$ shots



- As λ doubles, each time the line moves down but at a lower rate
- The slope of the line is fixed at $\frac{1}{2}$ for all λ

Applying CLT Mechanism

- Set $N = 2^8$ windows to get as straight a line as possible to accurately measure the intercept c
- 2^8 implies, for 2^{15} shots, the maximum window size is $w = 2^7$ shots giving us 6 data points $w = \{2^2, \ldots, 2^7\}$



- All intercepts were calculated as $c = \text{mean}(y - mx) \because m = \frac{1}{2}$

Expected Result

$$
\begin{aligned}
c &= y - mx && (1)\\
&= \log(\sigma/\mu) + \frac{1}{2}\log(w) && (2)\\
&= \log(\sqrt{wp(1-p)}/wp) + \frac{1}{2}\log(w) && (3)\\
&= \frac{1}{2}\log(\frac{1-p}{p}) = \frac{1}{2}\log(\frac{p_0}{p_1}) && (4)
\end{aligned}
$$

- Since $p \in (0,1)$, $c \in (-\infty, 0)$ as p increases, the line shifts down.
- Therefore we expect $c_2 - c_1 = \frac{1}{2}\log(\frac{1-p_1}{p_1}\frac{p_2}{1-p_2})$
- Or $c_2 - c_1 = \frac{1}{2}\log(\frac{1+p_{0\to1}-p_{1\to0}}{1+p_{1\to0}-p_{0\to1}})$
- All symmetric SPAM noise models should have $c = 0$.

Applying CLT Mechanism

- Readout Error $= (p_{0\to1} + p_{1\to0})/2$
- Readout Bias [6] $= p_{1\to0} - p_{0\to1}$ (Coming Down - Going Up)
- We can then use expected c values as $\frac{1}{2}\log(1 - \text{Readout Bias}/1 + \text{Readout Bias})$

Qubit	T_1	Readout Error	$p_{0\to1}$	$p_{1\to0}$	Readout Bias
61	232	0.0082	0.0099	0.0066	0.0033
129	232	0.0166	0.0186	0.0146	0.0040
5	232	0.0389	0.0350	0.0428	-0.0078

Qubit Parameters from IBM Torino's calibration

[6] Non-standard terminology

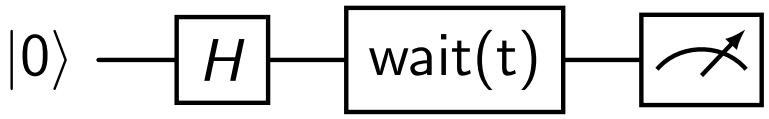
SPAM Results

Qubit	Expected c	Actual c
61	-0.0095	-0.125
129	-0.0115	-0.012
5	0.0225	0.033

- We're able to predict values of c within 0.01 of the actual value from just the calibration data
- Something went wrong with qubit 61.
- The possible reasons could be
  - Since it ran after two days in queue, the calibration data was outdated.
  - Single qubit gate error (Application of H), may have contaminated the results

T_1

- T_1 makes a qubit state decay over time as $P(|1\rangle) = e^{-t/T_1} p_1$ where t is the time since preparation.
- We will add Id^k before measurement for $Id^k = \{10^1, 10^2, 10^3\}$ of T_1.
- We will also vary for various values, subject to system constraints.
- Circuit used will be



Expected Result (Derivation)

- With $p_{1\to1} + p_{1\to0} = 1$, $p_{0\to0} + p_{0\to1} = 1$ and $\epsilon = 1 - e^{-t/T_1}$,
- And $p_0' = p_0 + p_1\epsilon$ and $p_1' = p_1 - p_1\epsilon$
- we can put these into the modified SPAM formula as
- $p_0^F = p_0' p_{0\to0} + p_1' p_{1\to0}$ and $p_1^F = p_0' p_{0\to1} + p_1' p_{1\to1}$
- Therefore

$$c = \frac{1}{2}\log\left(\frac{b(1-p_{0\to1}) + p_{1\to0}}{b(p_{0\to1}) + (1-p_{1\to0})}\right), \text{ where } b = \frac{1+\epsilon_1}{1-\epsilon_1} \tag{5}$$

[7]

We can see this formula very cleanly regresses to just SPAM noise when there's no noise.

[7] More generally $b = \frac{P(0)}{P(1)}$

Results

- Instead of listing all results we'll map it onto the processor architecture.
- We color code all results as $\Delta_\epsilon c = |c_{\text{pred}} - c_{\text{real}}|$

$\Delta_\epsilon c$	σ Ratio	Color	Meaning
< 0.5	< 1.41	green	Correct Prediction
$0.5 - 1.0$	$1.41 - 2$	yellow	Approximate, Other noise
> 1.0	> 2	black	Bad Qubit/Wrong Model

- Experiments are H gate followed by n of H, Id, X gates where $n \in \{10^1, 10^2, 10^3\}$.

Conclusions before the Diagram

- Bad qubits and bad connections show up reliably as errors
- Errors: Id results $<$ X results $\ll$ H results indicating single qubit gate noise in X, H
- H gate seems fundamentally different when being applied
- Short depth predictions are more reliable than long depth predictions

Results

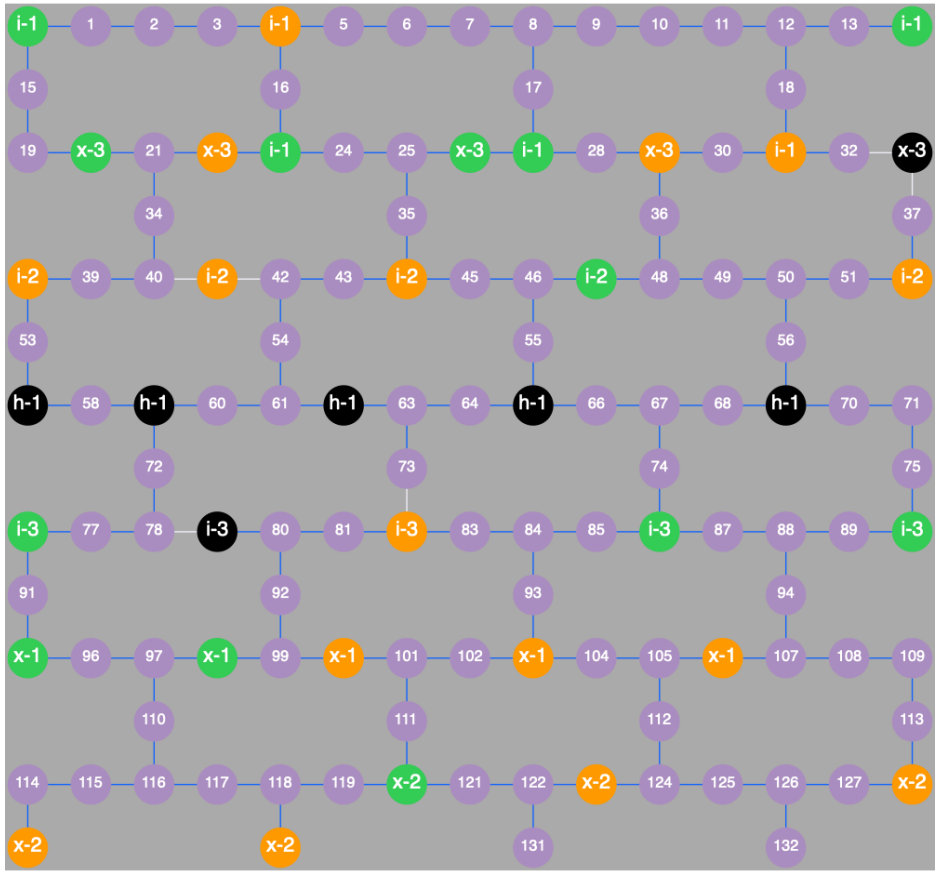


Figure: T1 results

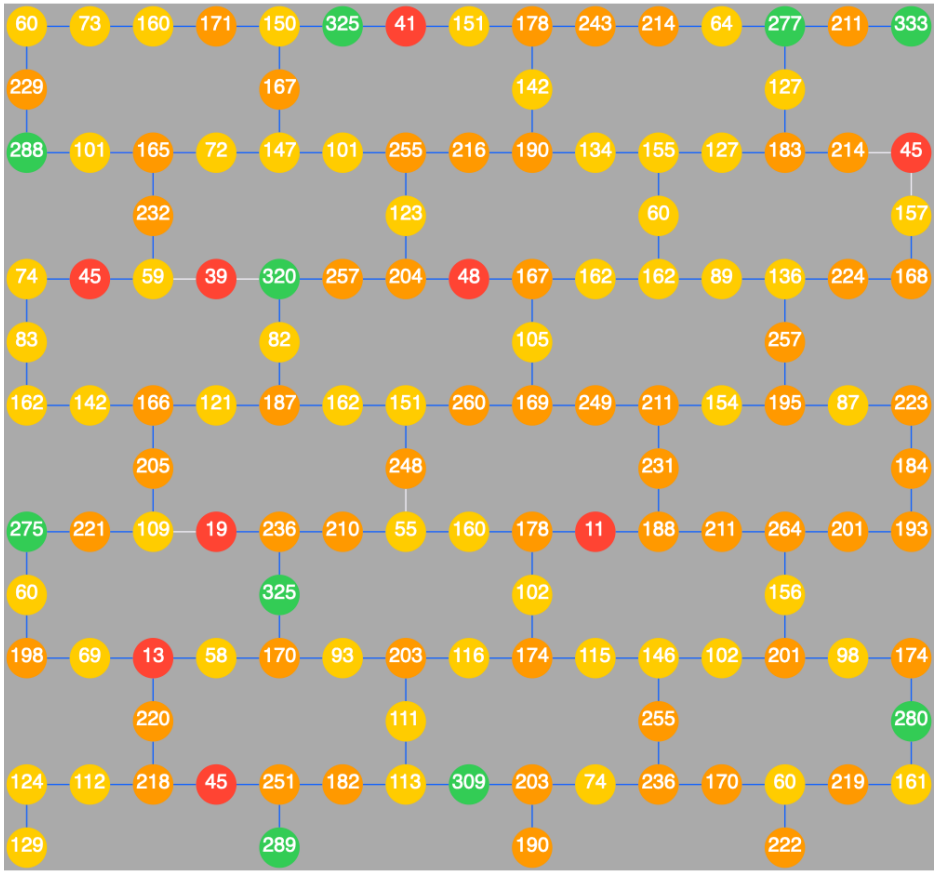


Figure: Torino T1 Map

Derivations: T_2

$$\rho_{T_1+T_2} = \begin{pmatrix} 1 + (|a|^2 - 1)\epsilon_1 & ab^*\epsilon_2 \\ a^*b\epsilon_2 & |b|^2\epsilon_1 \end{pmatrix} \because \epsilon_1 = e^{-t/T_1}, \epsilon_2 = e^{-t/T_2} \tag{6}$$

Multiplying by the H gate matrix, we get probabilities (no SPAM)

$$c = \frac{1}{2}\log\left(\frac{b(1 - p_{0\to1}) + p_{1\to0}}{b(p_{0\to1}) + (1 - p_{1\to0})}\right), \text{ where } b = \frac{1+\epsilon_2}{1-\epsilon_2} \tag{7}$$

- Formula modifies to above; **Assuming:** we measure in H basis
- We ran left half with extra measurements before circuit and right half without to check for correlated errors

Results: T_2

- Pre-measurements have no noticeable benefits in experiments
- There is no visible difference between single, double, and triple connected qubits in T_2 error
- Bad connections are not reliably worse as in T_1
- $\Delta_\epsilon c$ is both higher mean and higher variance than T_1 i.e we are worse at predicting T_2 than T_1
- Estimations are more accurate for high depth circuits than low depth circuits

Results

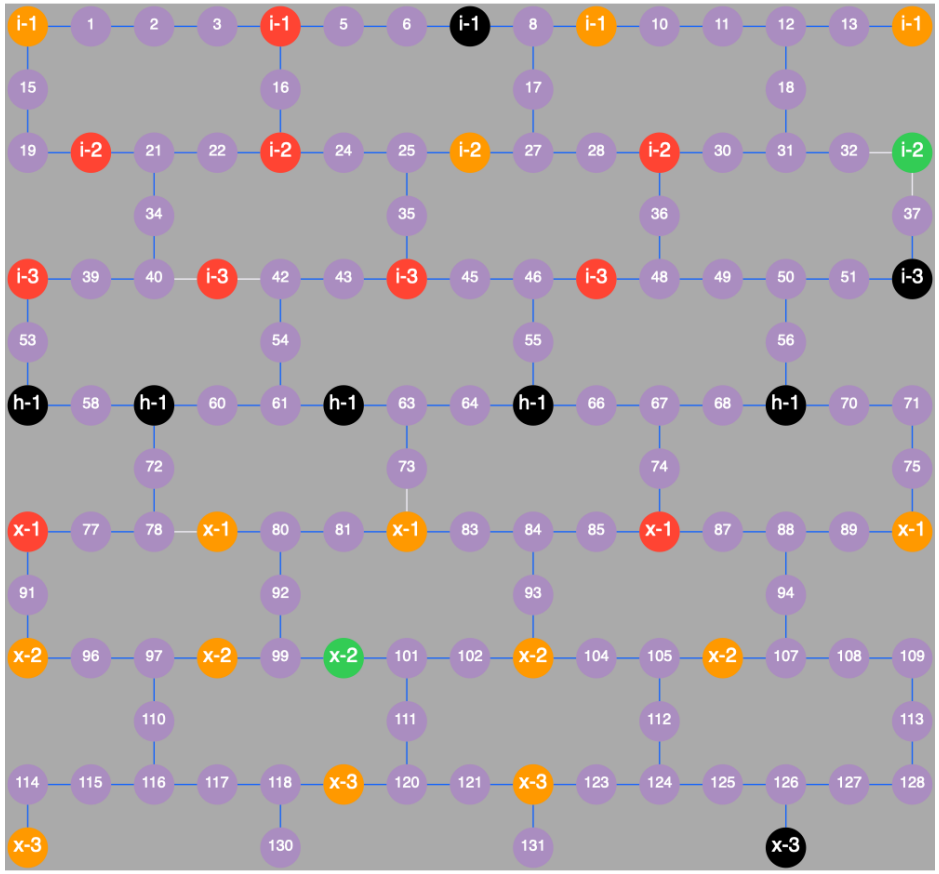


Figure: T2 results

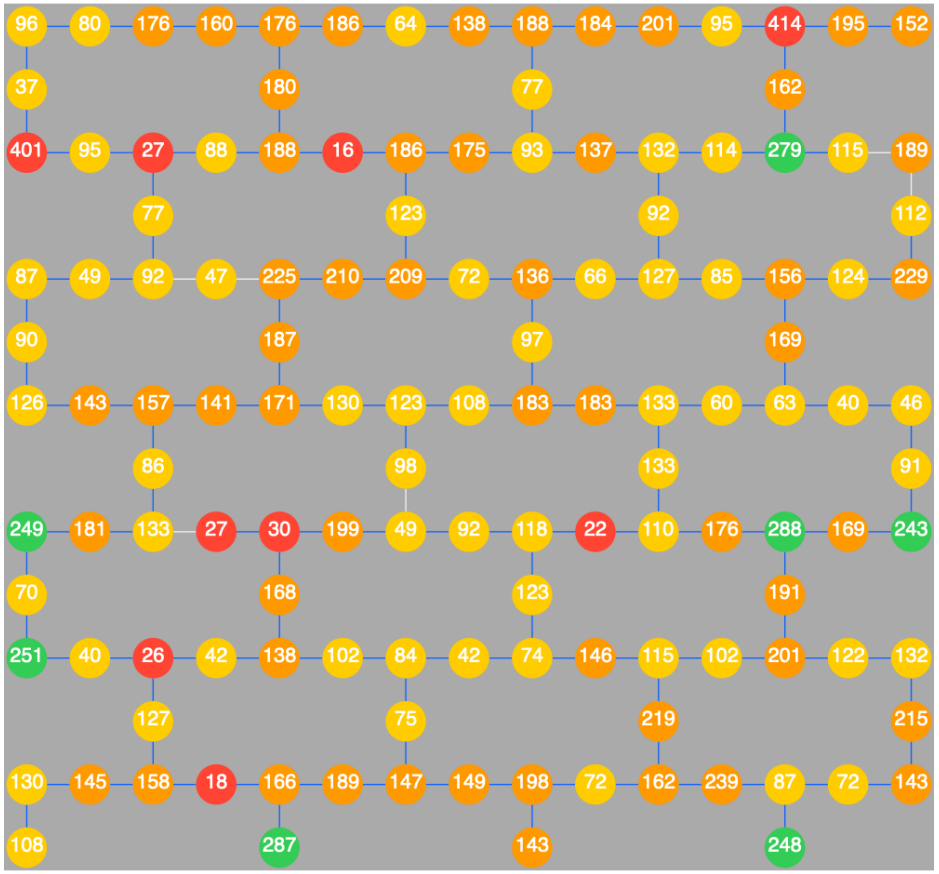


Figure: Torino T2 Map

Generalisation

- Having established our method works, and covered the major sources of noise, we will now generalise
- It is computationally intractible to calculate variance of an arbitrary Hamiltonian on an arbitrary state, precisely so we make a few assumptions:
  - No correlated errors and therefore all covar, coskewness etc terms are zero i.e all sources of noise are independent
  - Noise is independent of the circuit run and Hamiltonian evaluated
- Generally let our system Hamiltonian be H with some X_i sources of noise, then we can write

$$c = \frac{1}{2}\log\left(\frac{\max \mathrm{Var}(H) + \sum_i E[X_i^2] - \left(\sum_i E[X_i]\right)^2 + 2\langle H\rangle \sum_i E[X_i]}{\langle H\rangle}\right)$$

Correction

- We know max variance is an upper bound and will therefore have some error
- Let max $\mathrm{Var}(H) = Var_{\mathrm{true}}(H) + E_h$
- Via $log(a + b) = log(a) + log(1 + b/a)$, we can write our corrected c as
- $c = c_{\mathrm{true}} + log(1 + E_h/\mathrm{Var}(H_{\mathcal{E}}))$
- We can see for small E_h, $c = c_{\mathrm{true}}$

Application

- Our Hamiltonian has 15 terms
- Each term will add to the variance of the system Hamiltonian
- We treat n qubits as n independent systems and add the variances of the T_1s and T_2s
- Let approximate $\langle H \rangle = 1.84$
- So for a 4-qubit H_2 Hartree Fock circuit with 15 terms in the Hamiltonian, we would have

$$\mathrm{Var}(H_2) = \max \mathrm{Var}(H) + (4 \times \mathrm{Var}(T_1) + 4 \times \mathrm{Var}(T_2)) \times 15$$

We know what the system Hamiltonian for H_2 is and we can now apply it in each basis to get the expected c values.

Results: H_2

- It is known that variance of Hamiltonian only gives us an upper bound and assuming no large sources of error we can expect the actual variance to be lower which can be corrected using a constant offset

Runtime (ns)	T1 (μs)	T2 (μs)	Real	Pred	Error
1910	105	60	-2.820	-0.406	2.414
5060	112	108	-2.540	-0.391	2.149
1910	60	72	-2.970	-0.403	2.567
36560	150	160	-1.690	-0.042	1.648
5060	127	152	-2.460	-0.398	2.062

Results: H_2

- We estimate correction offset 2.1.
- To verify, we can test on a different machine (osaka) and see error after correction

Runtime (ns)	T1 (μs)	T2 (μs)	Real	Pred	Error
4900	140	58	-3.030	-2.472	0.558
36400	208	84	-1.590	-1.954	0.364
176400	268	99	-0.610	-1.242	0.632
4900	222	66	-2.070	-2.483	0.413

- As example, extrapolating first row $y = -0.5x - 2.472$
- We can now approximately predict shots for, here, the H_2 circuit
- at say 2^8 shots $\log(\frac{\sigma}{\mu}) = -6.472$ i.e $\frac{\sigma}{\mu} = 0.011$
- Hence 2^8 shots should have a $\sigma = 0.02$ (from $\mu = 1.84$)
- Real $\sigma = 0.012$ for 2^8 shots

Conclusions

- We have a shot prediction model which at worst is still accurate to within $\Delta_\epsilon c = 1$.
- As a conservative estimate we can always round up to the next power of 2
- 1 shot prediction may be a much more reliable way to estimate shots for large depth circuits

Known Limitations

- We have not accounted for correlated errors, crosstalk or gate errors
- It is unknown how H gate may add to error if applied many times on the same qubit as seen in T_1 results

Appendix: 1

For a stretched exponential distribution with normalised t, $\gamma_i = 1 - e^{-t^n}$, then it's p^{th} moments are:

$$\int_0^t t^p e^{-t^n} dt = \frac{1}{n}\gamma(\frac{p+1}{n}, t^n)$$

where γ is the lower incomplete Γ-function at circuit depth t. Now let's normalise and take a more real case.

- Normalised t is t/T_1 or t/T_2
- For most cases $n = 1$
- First and second moments (mean, variance) imply $p = 1, 2$

$$E_t(X) = 1 - (t+1)e^{-t} \tag{8}$$

$$E_t(X^2) = -t^2 e^{-t} + 2E_t(X) \tag{9}$$

Appendix: 2

Without loss of generality we may assume that H is real diagonal with eigenvalues λ_i. Then for some real value, $\lambda_i \in [\lambda_{\text{min}}, \lambda_{\text{max}}]$ which takes a value $(\frac{u_i}{||u_i||})^2$ with probability λ_i, we can write

$$\max \text{Var}(u) = \sum_i \lambda_i^2 u_i^2 - \left(\sum_i \lambda_i u_i \right)^2 = ||u||^2 \, \text{Var}(X)$$

Therefore trivially from Popoviciu's inequality for set X $(\sigma^2 \leq \frac{1}{4}(\sup(X) - \inf(X))^2)$, we can write

$$\text{Var}(X) \leq \left(\frac{\lambda_{\text{max}} - \lambda_{\text{min}}}{2} \right)^2$$

If the problem is not an eigenvalue finding problem, we can use a tighter bound from Bhatia-Davis inequality,
$\sigma^2 \leq (\lambda_{\text{max}} - \mu)(\mu - \lambda_{\text{min}})$.

Appendix: 3

Applying some CPTP noise channel ϵ to a state ρ,

$$\mathrm{Var}(H) = \langle\psi|H^2|\psi\rangle - \langle\psi|H|\psi\rangle^2 \tag{10}$$

$$\mathrm{Var}(H)_{\mathcal{E}} = Tr[H^2\mathcal{E}(\rho)] - Tr[H\mathcal{E}(\rho)]^2 \tag{11}$$

We can split the noise channel into multiple Kraus operators ϵ_k

$$\mathrm{Var}(H)_{\mathcal{E}} = \sum_k Tr[H^2 E_k \rho E_k^\dagger] - \left(\sum_k Tr[H E_k \rho E_k^\dagger]\right)^2 \tag{12}$$

$$= \sum_k Tr[E_k^\dagger H^2 E_k \rho] - \left(\sum_k Tr[E_k^\dagger H E_k \rho]\right)^2 \quad \because \text{Cycle Invariance} \tag{13}$$

Appendix: 3 (contd.)

Let $O_k = Tr[E_k^\dagger H E_k \rho]$, and $O_k^{(2)} = Tr[E_k^\dagger H^2 E_k \rho]$ then we can write

$$\mathrm{Var}(H)_{\mathcal{E}} = \sum_k O_k^{(2)} - \left(\sum_k O_k\right)^2 \tag{14}$$

$$= \langle H^2 \rangle + \sum_{k\neq 0} O_k^{(2)} - \left(\langle H \rangle + \sum_{k\neq 0} O_k\right)^2 \tag{15}$$

$$= \mathrm{Var}(H) + \sum_{k\neq 0} O_k^{(2)} - \left(\sum_{k\neq 0} O_k\right)^2 - 2\langle H \rangle \sum_{k\neq 0} O_k \tag{16}$$

Appendix: 3 (contd.)

For k in E_k let us define k as a bit-string of 0s and 1s, such that $E_{k_1 k_2 \ldots k_n} = E_{k_1} \otimes E_{k_2} \otimes \ldots \otimes E_{k_n}$, then we can write,

- E_0: $O(1)$ order noise
- E_1: $O(\gamma)$ order noise
- E_k: $O(\gamma^k)$ as higher order terms which we can ignore

This simplifies, for a given noise, assuming them to be identical for n qubits, to reduce down to $nO(\gamma)$ order noise terms and Identity. Therefore for, p sources of noise, we need only $pn + 1$ terms to estimate the variance of the system Hamiltonian.